Symmetry Method Homework 1

1) Time Reversal Symmetry

In physics, it is often said that the laws of physics is invariant under the time reversal transformation, i.e., T defined as

$$T(\text{equation}(t)) = \text{equation}(-t)$$

1.1) Show that the Newton's 3rd law of motion : $\vec{F} = m\vec{a}$ is invariant under $t \to -t$ action.

(Think of it as a symmetry action on an object)

1.2) Check that the set of symmetry actions $\{\mathbb{1}, T\}$ with a successive action as a multiplication is a group.

1.3) Find an isomorphism between the group $\{\mathbb{1}, T\}$ and $\mathbb{Z}_2$

2) A particle in a box in 3D

The eigenvalue of the Hamiltonian for a particle in 3D infinite square well (3D box) of equal length $L_x = L_y = L_z \equiv L$, is written as

$$E(n_x, n_y, n_z) = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

where n_x, n_y, n_z are principal quantum number for $\hat{x}, \hat{y}, \hat{z}$ directions. $(n_x, n_y, n_z = 0, 1, 2, \dots)$

- Notice that swapping $n_x \leftrightarrow n_y$ leaves E invariant:

$$E(n_y, n_x, n_z) = \frac{h^2}{8mL^2}(n_y^2 + n_x^2 + n_z^2) = E(n_x, n_y, n_z).$$

Therefore swapping $n_x \leftrightarrow n_y$ is a symmetry action of the energy eigenvalue.

- Also the cyclic swapping $n_x \to n_y \to n_z \to n_x$ leaves E invariant:

$$E(n_z, n_y, n_x) = \frac{h^2}{8mL^2}(n_z^2 + n_y^2 + n_x^2) = E(n_x, n_y, n_z)$$

This is also a symmetry action

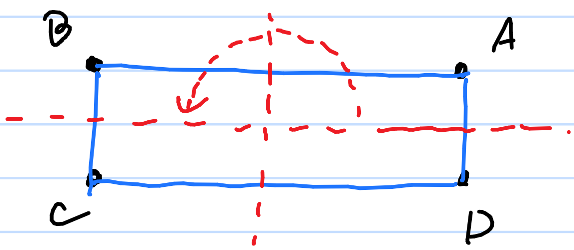
2.1) Find a symmetry action group containing actions mentioned above (it should look familiar to you)

2.2) What if the length of the box is not symmetric such that $L_x = L_y = L$ but $L_z \neq L_x$.

What is a symmetry action group now?

3) Dihedral group D_2

Consider the symmetry group of a rectangle



This group is called Dihedral group D_2 or Klein four group K_4

3.1) Find all elements and their multiplication table

3.2) Consider a group of 4 matrices

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$$

with the usual matrix multiplication.

Show that this group is isomorphic to D_2

4) Quaternion Group

Quaternion is a natural extension of complex numbers

The imaginaries are introduced $i^2 = j^2 = k^2 = ijk = -1$

and $jk = i$, $ij = k$, $ik = -j$, $kj = -i$, $ki = j$

$ji = -k$.

The Quaternion group Q_8 has 8 elements

$\{1, i, j, k, -1, -i, -j, -k\}$ with the usual multiplication.

4.1) Write down the multiplication table

4.2) Consider a group of 8 matrices

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \right.$$

$$\left. \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \right\}$$

with the usual multiplication.

Show that this group is isomorphic to Q_8